Band-1 Hadamard Order 4084 — Miyamoto (Paley-II seed H_1020); Gap 2052 Still OPEN

H_1020/C_1022/H_4084 PASS; 4084 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4036_Miyamoto`; `2026-09-15_Band1_Hadamard_3716_Miyamoto` (Miyamoto-only pattern).
Lab. `/workspace/experiments/band1/hadamard_4084_miyamoto/`.
Cutoff. 2026-09-15T14:05:56Z (~10:05AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1020 and 4084 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{1020} ($q = 509$), and documents an alternate Paley-I seed ($q = 1019$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4084 is a **Miyamoto-only bonus (NOT in SPK)**: Paley-I **FAIL** ($q = 4083 = 3 \cdot 1361$ not prime); Paley-II **FAIL** ($q = 2041 = 13 \cdot 157$ not prime); Kronecker **FAIL** ($4084 = 4 \cdot 1021$, 1021 not Hadamard-admissible). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4036 on box. This increment ships **order 4084** ($4084 = 4 \cdot 1021$, 1021 prime $\equiv 1 \pmod 4$) via Miyamoto with a direct **Paley-II seed** H_{1020} ($q = 509$). Symmetric Paley conference C_{1022} over prime 1021; $H_{4084} = \text{Miyamoto}(H_{1020}, C_{1022})$. Alternate seed H_{1020} via Paley type I ($q = 1019$) also verified. Unit tests **29/29 PASS**. SPK next gap remains **2052** (OPEN). Order 4084 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4084; Miyamoto; Paley type II; Paley type I; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{1020} constructed and verified (Paley type II, $q = 509$). 2. H_{1020} also constructed and verified (Paley type I, $q = 1019$) — alternate seed. 3. C_{1022} constructed and verified (Paley conference, $q = 1021$): $CC^\top = 1021I$, zero diagonal. 4. H_{4084} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $4084 \notin$ SPK (Miyamoto-only). Paley-I/II/Kronecker paths for 4084 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$4084 = 4 \cdot 1021$
$q = 1021$	prime, $1021 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1020}
Needed conference	order $1022 = q + 1$
H_{1020} path (used)	Paley type II, $q = 509$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 1020$
H_{1020} alternate	Paley type I, $q = 1019$ (prime $\equiv 3 \pmod 4$); $q + 1 = 1020$
Conference path	Symmetric Paley over prime 1021
Paley-I for 4084	**REJECTED**: $q' = 4083 = 3 \cdot 1361$ not prime
Paley-II for 4084	**REJECTED**: $q' = 2041 = 13 \cdot 157$ not prime
Kronecker for 4084	**REJECTED**: only 4×1021; $1021 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{1020}	Paley type II, $q = 509$	0	0.6436	**PASS**
H_{1020} alt	Paley type I, $q = 1019$	0	1.0515	**PASS**
C_{1022}	Paley conference, $q = 1021$	0	1.0368	**PASS**
H_{4084}	Miyamoto(H_{1020}, C_{1022})	0	29.7106	**PASS**

Primary pipeline wall (incl. unit tests + alternate seed + SPK refresh) ≈ 33.3694 s. Compressed ± 1 matrices saved as `H1020_pm1.npz`, `H1020_paleyI_alt_pm1.npz`, `C1022_pm1.npz`, `H4084_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior classical/SPK ships (through 4036/3908/3748/…; **exclude** 3716/3524/3284/3076/3044/2836/2692/2564/**4084** Miyamoto-only) $\cup$ verified H_{1020} (Paley-II) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4500). **Miyamoto is not folded into SPK**; order 4084 remains a Miyamoto-only bonus.

Quantity	Value
Known count	964
4084 in SPK	**False** (Miyamoto-only)
1020 in SPK	**True** (Paley-II)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**Yes** (like 3716/3524/…)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

29/29 PASS (Sylvester/Paley/Miyamoto controls; $q = 509/1019/1021$ arithmetic; Paley-II seed for H_{1020}; SPK path rejection for 4084; Miyamoto-only arith guard; reject-all-ones; conference

control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 4084 is a **Miyamoto-only bonus (NOT in SPK)**.

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 1033 \to n = 4132$ PaleyI(q=1031)) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_4084_miyamoto/` · *Script:* `verify_hadamard_4084.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.